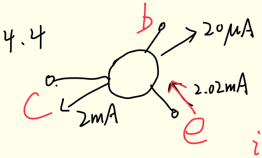
4.4

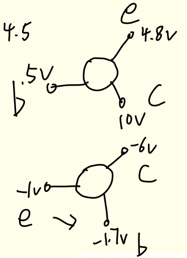

b
20μA
2.02mA
C
2mA
e

$i_E > i_C > i_B$

$\beta = \frac{i_C}{i_E} = \frac{2mA}{2\mu A} = 100.$

i_E流入，为NPN型

4.5

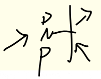

e
4.8V
5V
b
C
10V
-6V
C
-1V
e
-1.7V
b

由5−4.8 = 0.2判定为锗管

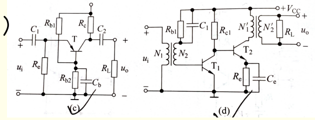
C区电位最高，NPN型

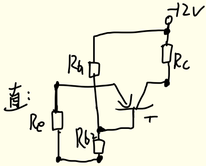
−1.7−(−1) = 0.7 判为硅管.

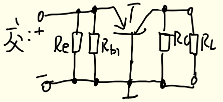
PNP型



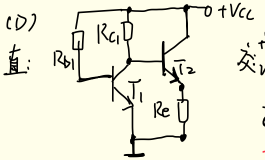
4.9. (c)

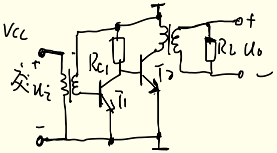

−12V
R_{b1}
R_c
C_1
C_2
T
R_e
R_{b2}
C_b
u_i
u_o
(c)
$+V_{CC}$
R_{b1}
C_1
R_{c1}
N_1'
N_2'
R_L
u_o
N_1
N_2
T_1
T_2
R_e
C_e
(d)

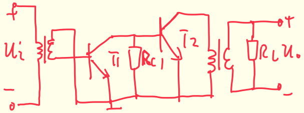

-12V
直:
Rb
Rc
Re
Rb2
T

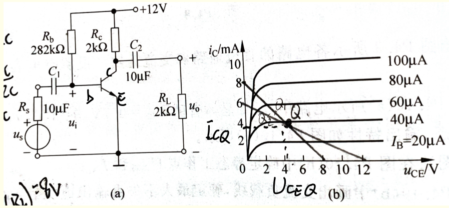

交:
+
Re
Rb1
T
Rc
RL

(D)


+Vcc
直:
Rb1
Rc1
T1
T2
Re


交
ui
Rc1
T1
T2
+
RL
Uo


+
Ui
T1
Rc1
T2
RL
U0
+
-


+12V
R_b
282kΩ
R_c
2kΩ
C_2
10μF
C_1
10μF
R_s
u_s
u_i
R_L
2kΩ
u_o
(a)
(RL)=8V
i_C/mA
10
8
6
4
2
0
2
4
6
8
10
12
u_{CE}/V
100μA
80μA
60μA
40μA
I_B=20μA
Q
ICQ
UCEQ
(b)

4.10

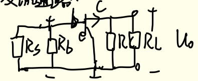
(1) 由KVL: $12 - R_c \cdot I_{CQ} = U_{CEQ}$ ①

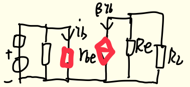
$12 - I_{BQ}R_b = U_{CEQ}$ ② $\Rightarrow I_{BQ} \approx 40\mu A$.

可得直流负载曲线，与$I_B = 40\mu A$交于Q为静态工作点

得$U_{CEQ} \approx 4V$. $I_{CQ} \approx 4mA$

(2) 交流通路:


C
b
e
Rs
Rb
Rc
RL
Uo


βib
ib
+
-
rbe
Re
RL

$V_{CC}' = U_{CEQ} + I_{CQ}R_L'$

$= U_{CEQ} + I_{CQ}(R_c // R_L)$

$= 8V$

$U_{om} = \frac{4 - U_{BE}}{\sqrt{2}} \approx 2.33V$

(3)

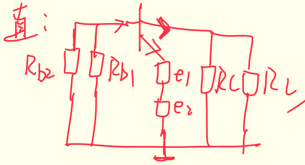
$V_{CC}' - U_{CEQ} = 4V > U_{CEQ} - U_{BE} = 3.3V.$

饱和

(4). $R_b \downarrow$ → Q左移 → $U_{om} \downarrow$

$R_c \uparrow$ → Q左移. → $U_{om} \downarrow$

$R_L \uparrow$ → Q, U_{om} 不变


直:
Rb2
Rb1
e1
e2
RC
RL

4.12

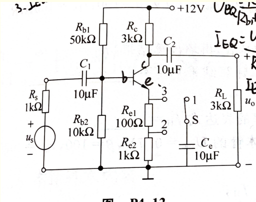

+12V
R_{b1}
50kΩ
R_c
3kΩ
C_2
10μF
C_1
10μF
b
c
e
R_s
1kΩ
u_s
R_{b2}
10kΩ
R_{e1}
100Ω
R_{e2}
1kΩ
3
2
1
S
C_e
10μF
R_L
3kΩ
u_o

图 P4.12

$U_{BQ} = \frac{R_{b2}}{R_{b1}+R_{b2}} V_{CC} = 2V$

$I_{EQ} = \frac{U_{BQ} - U_{BE}}{R_{e1}+R_{e2}} \approx I_{CQ} \approx 1.182mA$

$I_{BQ} = \frac{I_{CQ}}{\beta} \approx 3.94\mu A$

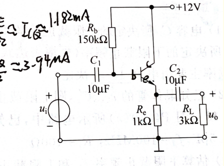

+12V
R_b
150kΩ
C_1
10μF
b
c
e
C_2
10μF
u_i
R_e
1kΩ
R_L
3kΩ
u_o

图 P4.13

4.11 (1) $I_{CQ} \approx 4mA$, $r_{bb'} + \beta\frac{U_T}{I_{CQ}} = (100 + 100 \times \frac{26}{4})\Omega = 750\Omega$

$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_i} = \frac{\beta(R_c // R_L)}{r_{be}} \approx -133.3$, $R_i = R_b // r_{be} \approx 750\Omega$

$\dot{A}_{us} = \frac{\dot{U}_o}{\dot{U}_s} = \frac{\dot{U}_i}{\dot{U}_s} \cdot \frac{\dot{U}_o}{\dot{U}_i} = \frac{R_i}{R_i + R_s} \cdot \dot{A}_u \approx -57$, $R_o = R_c = 2k\Omega$

(2) $R_b \uparrow$ → $I_{BQ} \downarrow$ → $r_{be} \uparrow$ → $|\dot{A}_u| \downarrow$ → $R_i \uparrow$、R_o不变.

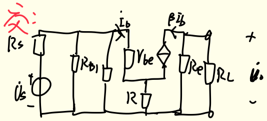
$R_s \uparrow$ → $\dot{A}_u$、R_i、R_o不变，$|\dot{A}_{us}| \uparrow$

$R_c \uparrow$ → R_i不变，$|\dot{A}_u| \uparrow$、$|\dot{A}_{us}| \uparrow$，$R_o \uparrow$

$R_L \uparrow$ → R_i、R_o不变，$|\dot{A}_u| \uparrow$，$|\dot{A}_{us}| \uparrow$

4.12.

(1) 设$V_{CC} = 12V$.


交:
Rs
$\dot{I}_b$
$\beta\dot{I}_b$
+
$\dot{U}_s$
Rb1
rbe
R
Re
RL
$\dot{U}_o$
-

$U_{BQ} = \frac{R_{b2}}{R_{b1}+R_{b2}} \cdot V_{CC} = 2V$

$I_{BQ} = \frac{I_{CQ}}{\beta} \approx 3.94\mu A.$

$I_{EQ} = \frac{U_{BQ} - U_{BE}}{R_{e1}+R_{e2}} \approx 1.182mA \approx I_{CQ}$

$U_{CEQ} = V_{CC} - I_{CQ}R_c - I_{EQ}(R_{e1}+R_{e2}) \approx 7.15V$

(2) $r_{be} = r_{bb'} + \beta\frac{U_T}{I_{CQ}} = (200 + 300 \times \frac{26}{1.182}) = 6799\Omega$

$\dot{A}_1 = \frac{\dot{U}_o}{\dot{U}_i} = -\frac{\beta(R_c // R_L)}{r_{be} + (1+\beta)(R_{e1}+R_{e2})}$ ①

$\dot{A}_2 = -\frac{\beta(R_c // R_L)}{r_{be} + (1+\beta)R_{e1}}$ ②

$\dot{A}_3 = \frac{\beta(R_c // R_L)}{r_{be}}$ ③

$\Rightarrow \dot{A}_1 \approx -1.33$

$\dot{A}_2 \approx -12.2$

$\dot{A}_3 \approx -66.2$

4.13.

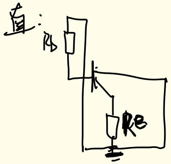
(1) $I_{BQ}=\frac{12-U_{BE}}{R_b+(1+\beta)R_e}$ ①

$I_{CQ}=\beta I_{BQ}$ ②

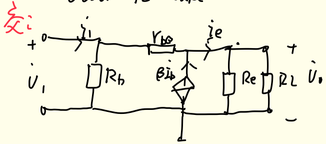
$U_{CEQ}=12-I_{CQ}R_e$ ③

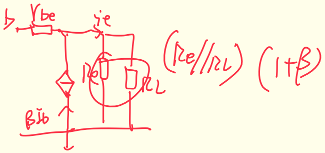
$\Rightarrow\begin{cases} I_{BQ}\approx 28.2\mu A \\ I_{CQ}\approx 7.05mA \\ U_{CEQ}\approx 4.95V. \end{cases}$

直:



交:


+
$\dot{U}_i$
$\dot{I}_i$
Rb
r_{be}
$\beta\dot{I}_b$
$\dot{I}_e$
Re
RL
+
$\dot{U}_o$
−


b
r_{be}
$\dot{I}_e$
Re
RL
$\beta\dot{I}_b$
$(Re//RL)(1+\beta)$

$r_{be}=r_{bb'}+\beta\frac{U_T}{I_{CQ}}\approx 1222\Omega$

$\dot{A}_u=\frac{\dot{U}_o}{\dot{U}_i}=\frac{(1+\beta)(Re//R_L)}{r_{be}+(1+\beta)(Re//R_L)}\approx 0.99.$

$R_i=R_b//[r_{be}+(1+\beta)(Re//R_L)]\approx 83.7K\Omega$

$R_o=Re//\frac{r_{be}}{1+\beta}\approx 4.86\Omega.$

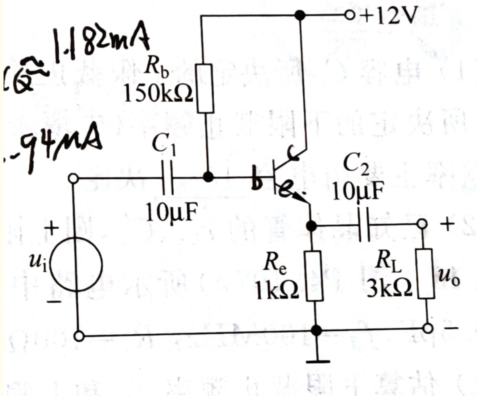

1.182mA
IQ≈
-94mA
+12V
R_b
150kΩ
C_1
10μF
b
c
e
C_2
10μF
+
u_i
−
R_e
1kΩ
R_L
3kΩ
+
u_o
−

图 P4.13

电路微变分析，等效转化为对应Rbe电阻和电流源(受控)很重要，画好直流通路和交流通路，利用戴维宁等效能方便求出Ri，Ro，找到静态工作点，利于分析当某个元件变换时，其它元件的变化情况.

202410711129 邓可